

Arumugam N

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Summary

Computer Science graduate with strong foundation in full-stack web development, database management, and enterprise application development. Proven experience in building RESTful web services, working with SQL databases, and delivering production-ready applications using Agile methodologies. Demonstrated ability in web technologies, API development, and system integration. Seeking Java Developer or Software Engineer roles to leverage technical expertise in web application development and database technologies.

Experience

Python Full Stack Developer

January 2025 – May 2025

Qspiders Campus Connect

Chennai, Tamil Nadu

- Engineered 5+ enterprise web applications using Python frameworks with MySQL database integration, implementing MVC architecture patterns and serving business requirements for 1000+ users
- Developed and deployed RESTful web services with JSON/XML data exchange, ensuring seamless frontend-backend integration and reducing API response times by 30%
- Implemented database connectivity using JDBC principles, optimized SQL queries, and followed SDLC best practices with Git version control in Linux development environment
- Applied Agile methodologies and participated in sprint planning, daily standups, and code reviews, consistently delivering features ahead of project deadlines

Software Development Intern

January 2024 – March 2024

Quantanics Private Limited

Madurai, Tamil Nadu

- Developed web-based ML application with backend services, implementing RESTful API architecture and achieving 95% system uptime with efficient database queries
- Built enterprise-grade software solutions following Agile development cycles, participating in requirements analysis, system design, and code deployment phases
- Collaborated with cross-functional teams using Scrum methodology, maintaining code quality through unit testing and version control, consistently meeting sprint deadlines
- Gained hands-on experience with Linux systems, database connectivity, and web service integration for scalable application development

Technical Skills

Programming Languages: Python, C, C++, SQL, JavaScript, HTML5, CSS3

Machine Learning: TensorFlow, PyTorch, OpenCV, YOLOv8, Scikit-learn, NLP, Generative AI, LLMs, Computer Vision

Web Development: Django, Flask, RESTful APIs, MongoDB, MySQL, Frontend Integration

Tools and Platforms: n8n, Git, VS Code, PyCharm, Eclipse, Streamlit, JupyterLab, NVIDIA Jetson

Data Analysis: Pandas, NumPy, Matplotlib, Statistical Modeling, Data Visualization

Methodologies: Agile, Scrum, SDLC, Version Control, Unit Testing, Debugging

Projects

podCaster: AI-Powered Podcast Generator | *Django, Python, AI, HTML, CSS, JavaScript* June 2025 – July 2025

- Built full-stack web application that generates podcast scripts and simulates AI-driven conversations using DeepSeek R1, streamlining content creation workflow for podcasters
- Designed intuitive two-step user interface with topic/title input phase and real-time chatbot interaction flow, improving user experience and engagement
- Integrated advanced prompt engineering techniques to generate coherent, engaging scripts based on user input, ensuring high-quality content output
- Implemented optional text-to-speech conversion using local Piper TTS engine, transforming markdown scripts to voice format for immediate podcast production

MedBot: AI-Integrated Rover | *Python, TensorFlow, Jetson SDK, OpenCV*

Jan 2023 – Mar 2023

- Led development of autonomous robotics solution using NVIDIA Jetson Nano for identifying medicinal plants in remote areas, reducing manual identification time by 80%
- Implemented deep learning image classification models achieving 92% accuracy in plant species differentiation based on leaf pattern analysis across 50+ plant varieties

- Integrated real-time computer vision pipeline processing 30 FPS video feed with optimized algorithms for field deployment in challenging outdoor environments
- Published patent: "MedBot: An AI-integrated Rover for Medicinal Plant Detection" (January 2024) - recognized for innovation in agricultural technology

Org-Assist: Intelligent Organizational Chatbot | *Python, MongoDB, NLP, Flask*

Oct 2023 – Nov 2023

- Built full-stack web application featuring intelligent chatbot for organizational information and complaint management, handling 500+ user queries with 90% satisfaction rate
- Deployed advanced NLP techniques using Python libraries for natural language understanding and context-aware response generation, reducing response time by 60%
- Integrated MongoDB database with Flask backend for scalable data storage and retrieval, supporting concurrent user sessions

Education

Francis Xavier Engineering College

Bachelor of Technology in Computer Science and Business Systems

CGPA: 8.6/10.0

Tirunelveli, Tamil Nadu

Sept 2021 – Apr 2025

Tilak Vidhyalaya Higher Secondary School

Higher Secondary Certificate

Percentage: 91.5%

Tirunelveli, Tamil Nadu

June 2019 – May 2021

Achievements Certifications

First Prize – Srishti International Project Exhibition for MedBot innovation (March 2023)

First Prize – Internal Hackathon, Francis Xavier Engineering College, Software Domain (October 2023)

Patent Publication – "MedBot: An AI-integrated Rover for Medicinal Plant Detection" (January 2024)

NPTEL Certification – Natural Language Processing Fundamentals with hands-on Python implementation (October 2023)

Oracle Certification – Certified Professional of Generative AI (August 2025)